

Iterative Design and the Future of Flyber



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Step 1

KPIs and Prior Experiment

Measure completed demand, not clicks alone

- North-star KPI: ride-launch conversion per opened session = sessions with begin_ride / sessions with open. It connects customer intent to Flyber's transaction model and is directly calculable from the logs.

KPI	Why it matters	Available calculation
Ride-launch conversion	Completed demand and revenue opportunity	begin_ride sessions / open sessions
Search completion	Diagnoses booking-form friction	search sessions / #_of_users sessions
Ride frequency	Repeat engagement	begin_ride sessions / unique user IDs
Not available	Revenue, margin, retention, safety	Need fare/cost, cohort date, cancellation and incident data

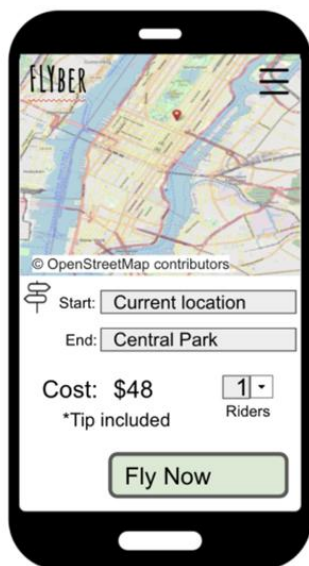
The first test changed CTA wording and price detail

- Control: 'Book Flight' + tip included. Exp 1: 'Fly Now' + tip included. Exp 2: 'Book Flight' with tip text removed. Exp 3: 'Fly Now' with tip text removed. The 2 x 2 structure tests CTA wording and price transparency, but the unofficial implementation reused many users across variants.

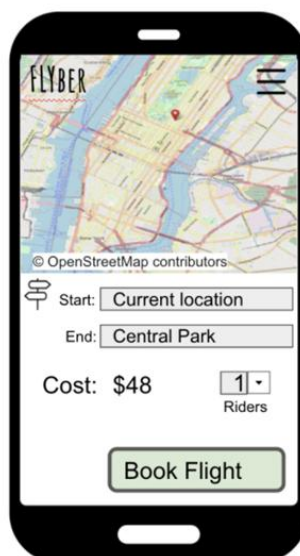
Control



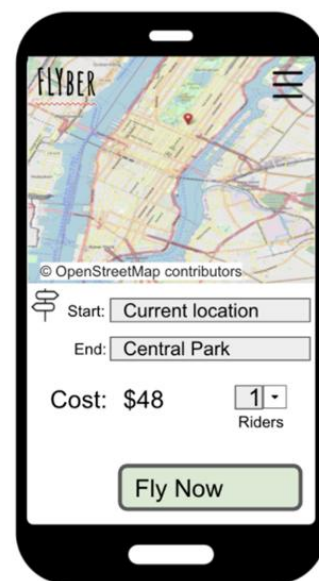
Experiment 1



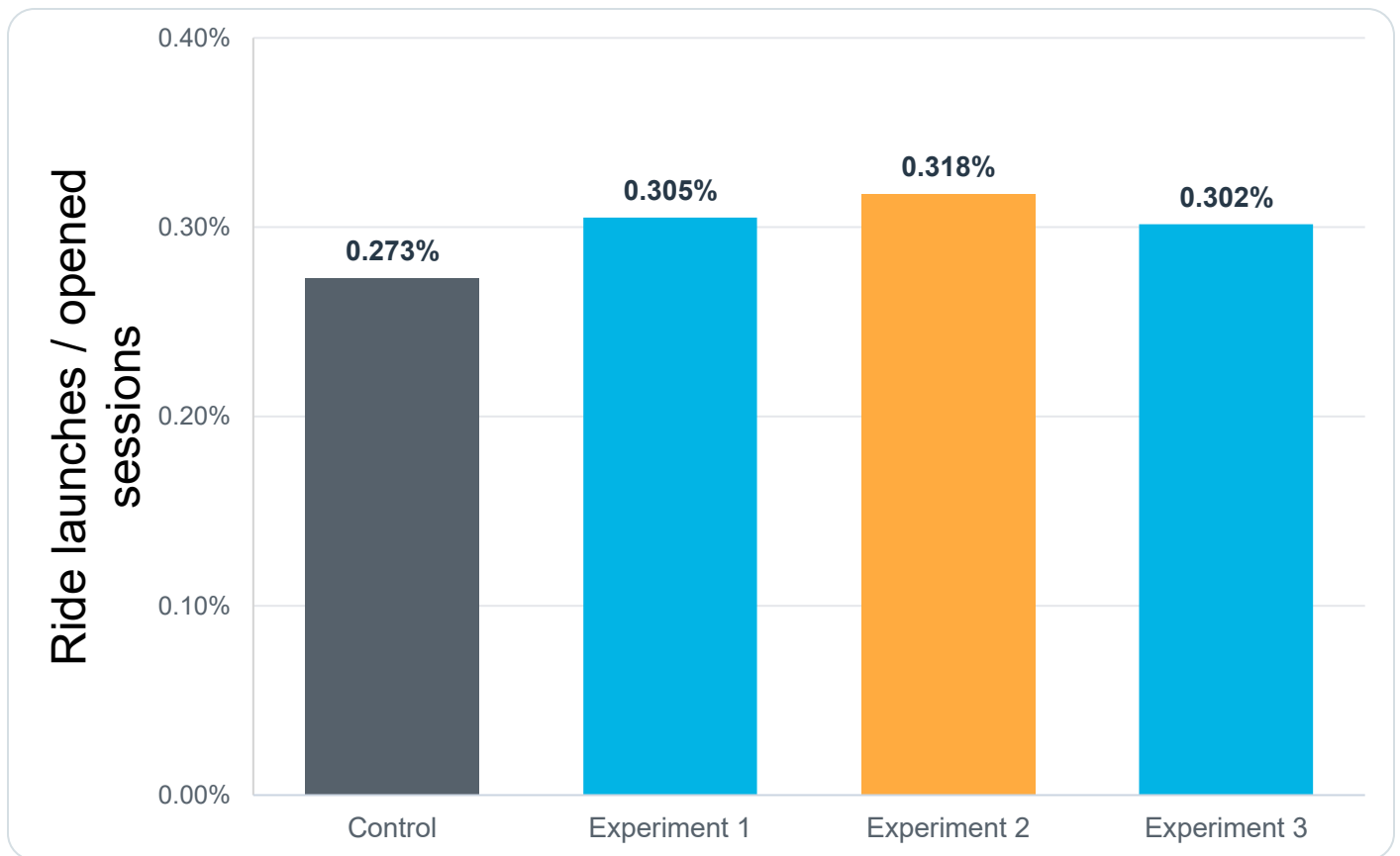
Experiment 2



Experiment 3



Prior variants did not reliably improve conversion



n = 56,390 to 56,688 sessions per condition

- Experiment 2 was directionally best at 0.318% ride launches per opened session, versus 0.273% for control. Its +0.044 percentage-point lift is small and uncertain.

The prior experiment is inconclusive

A Welch two-sample t-test is appropriate because it compares mean binary ride outcomes (conversion rates) between two independent groups without assuming equal variances. All Holm-adjusted p-values are 0.506-0.636 (> 0.05).

Variant	Rate	Abs. lift	95% CI of lift	Holm p
Experiment 1	0.305%	+0.032 pp	-0.031 to +0.095 pp	0.636
Experiment 2	0.318%	+0.044 pp	-0.019 to +0.108 pp	0.506
Experiment 3	0.302%	+0.029 pp	-0.034 to +0.091 pp	0.636

Decision: do not expand any experiment. None is statistically significant (all Holm-adjusted p-values > 0.05). Retain control and rerun a pre-registered, user-randomized test.

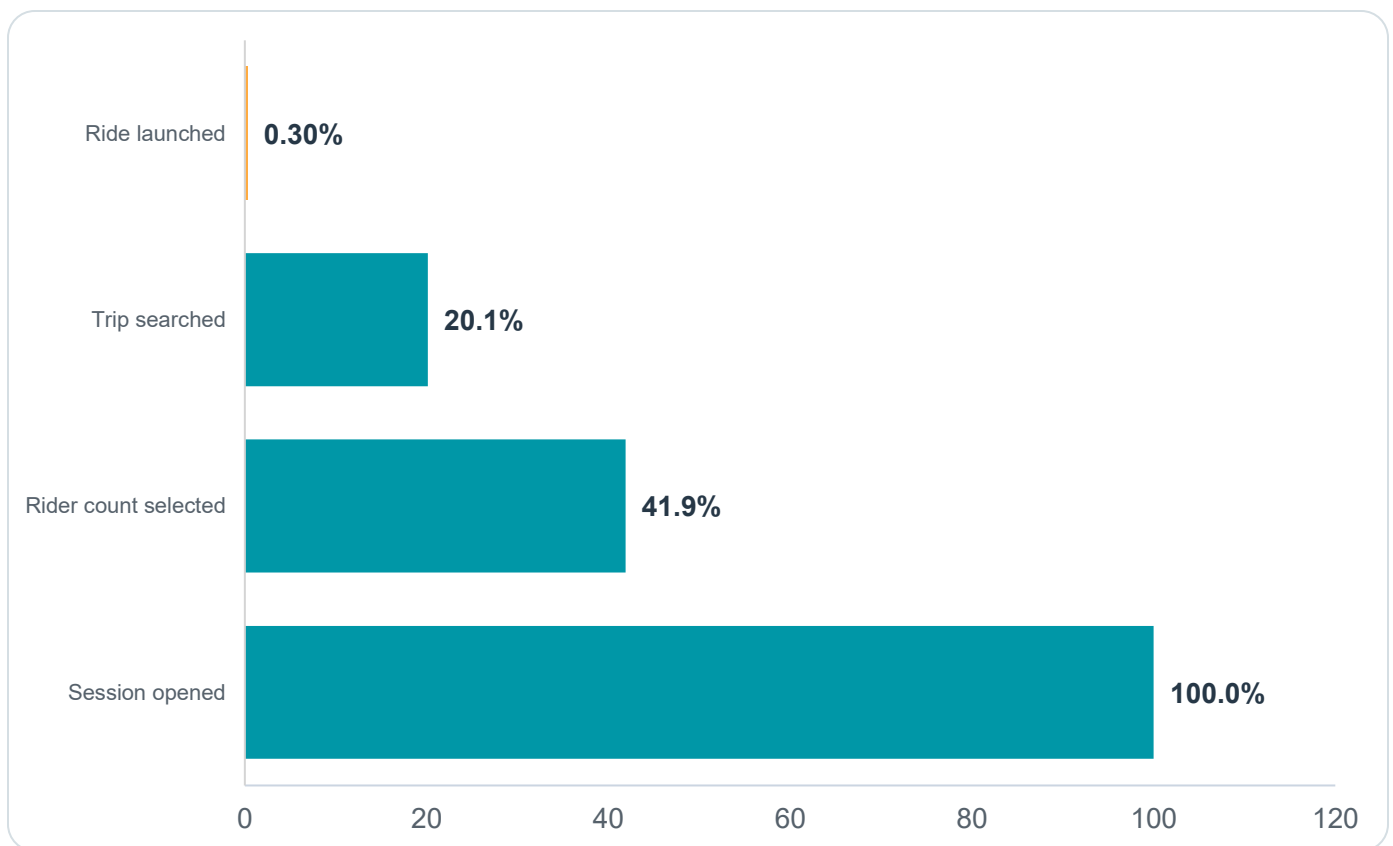


Step 2

Funnel and Cohort Analysis

Only 0.30% of sessions launch a ride

Four stages are strictly ordered: open -> rider count -> search -> begin ride. Search to ride has the largest relative drop (98.5%); open to rider count loses the most sessions (131,407).



Stage counts: 226,155 -> 94,748 -> 45,503 -> 677

Age and neighborhood define the cohorts

- Age 50+ is the largest age category (64,059 user IDs appearing in that category); Manhattan is the largest stable neighborhood cohort (57,110 users). Age is inconsistent across repeated user IDs, so age results are interpreted at the session-profile level.

Age	User IDs*	Sessions
18-29	28,321	33,878
30-39	20,124	22,760
40-49	41,774	56,256
50+	64,059	113,261

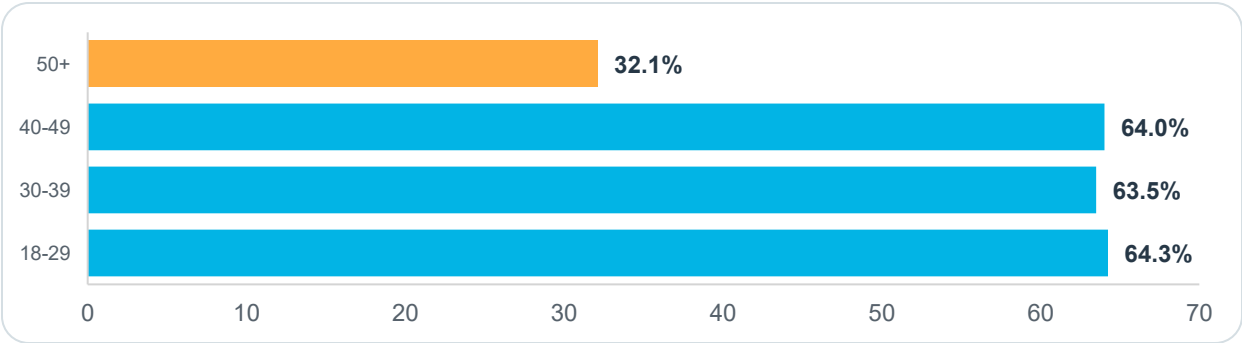
Neighborhood	Users	Sessions
Manhattan	57,110	158,366
Brooklyn	16,435	45,583
Queens	4,050	11,172
Bronx	2,396	6,693
Staten Island	1,566	4,341

*A user ID can appear in more than one age category; neighborhood is stable per user.

Age 50+ users abandon before search

After rider-count selection, 32.1% of age-50-plus sessions reach search versus 64.0% under age 50 ($p < 0.001$). Their 0.172% end-to-end rate is 42.5% below average.

Age	Open -> select	Select -> search	Search -> ride	Open -> ride
18-29	42.0%	64.3%	1.38%	0.372%
30-39	41.6%	63.5%	1.66%	0.439%
40-49	41.9%	64.0%	1.70%	0.455%
50+	41.9%	32.1%	1.28%	0.172%



Neighborhood conversion is uniform

All five neighborhoods follow nearly the same funnel. Select-to-search ranges from 47.3% to 48.8%; end-to-end conversion ranges from 0.278% to 0.344%. Age is the larger gap.

Neighborhood	Sessions	Open -> select	Select -> search	Search -> ride	Open -> ride
Manhattan	158,366	42.0%	48.1%	1.48%	0.299%
Brooklyn	45,583	41.8%	47.8%	1.48%	0.296%
Queens	11,172	41.4%	48.8%	1.37%	0.278%
Bronx	6,693	41.2%	48.2%	1.73%	0.344%
Staten Island	4,341	42.2%	47.3%	1.61%	0.323%

Recommendation: prioritize the age-related friction, while collecting destination, device, accessibility-setting and pricing-response data for richer future cohorts.



Step 3

Hypothesis and Next
Steps

Older users report accessibility friction

- The break occurs after rider-count selection. Older interviewees independently describe small controls, unreadable instructions and absent voice support while still valuing Flyber's speed.

"I let my assistant actually book... because... the instructions were too small."
- Louis Jones, age 70

"I prefer to use text-to-speech which I can't do with Flyber."
- Kierran Blackburn, age 55

"I don't need to fiddle around with an app and hitting tiny buttons."
- Charlie Johnson, age 75

Dataset note: the supplied interview file contains 11 respondents, not the 20 stated in the overview.

Make destination entry easier for age-50-plus users

- Hypothesis: Because age-50-plus sessions convert from rider selection to search at 32.1%, versus about 64% for younger groups, reducing visual and input friction for age-50-plus users will raise this step by at least 3 percentage points (to 35.1% or more).

Condition	Feature concept	User need addressed
Control	Current booking flow	Baseline
Variant A	Larger type, controls and contrast	Readable, easier targeting
Variant B	Voice-assisted destination entry	Hands-free, lower typing burden
Variant C	Saved and recent destinations	Fewer repeated entry steps

Run a user-randomized four-arm test for at least seven days

- Expose only age-50-plus users after app open; persist assignment by user ID across sessions. Allocate 25% to control and each variant. Primary outcome: search within the same session after rider-count selection.

Design element	Specification
Primary metric	$\text{select_to_search} = \text{search} / \text{rider_count_selected}$
Secondary	open-to-ride, search-to-ride, completion time
Guardrails	error rate, abandonment, latency, cancellation, support contact
Decision	+3 pp MDE; 80% power; 0.0167 alpha per comparison
Sample/runtime	5,192 eligible sessions/arm; 20,768 total; minimum 7 days

Observed eligible volume implies about 3.1 days to reach sample size; seven days is retained to cover a complete weekly cycle.



Appendix

Methods and Evidence

Data quality limits causal claims

The log is structurally complete, but the experiment does not preserve user-level assignment and age changes across repeated user IDs. Results are therefore session-level and do not justify causal expansion.

Check	Result	Implication
Event rows	367,083	Full supplied log
Missing cells / duplicate IDs	0 / 0	No row-level repair needed
Sessions in multiple variants	0	Session assignment is internally consistent
Users in multiple variants	46,894	Cross-session contamination risk
Users with multiple ages	52,576	Treat age as session-profile data
Users with multiple neighborhoods	0	Neighborhood is stable

Detailed experiment statistics

Unit: opened session; outcome: at least one begin_ride event. Welch's t-test is appropriate because it compares mean binary outcomes (conversion rates) between independent groups without assuming equal variances.

Group	Sessions	Rides	Rate	Rel. lift	t	Raw p	Holm p
Control	56,390	154	0.273%	-	-	-	-
Experiment 1	56,390	172	0.305%	+11.7%	0.998	0.318	0.636
Experiment 2	56,688	180	0.318%	+16.3%	1.377	0.169	0.506
Experiment 3	56,687	171	0.302%	+10.5%	0.897	0.370	0.636

All 95% CIs cross zero and all Holm-adjusted p-values exceed 0.05. No experiment is statistically significant; do not expand any experiment.

Instrument the next test before launch

The four existing events support the funnel. The next test also needs exposure, interaction, error and accessibility properties so each KPI has an auditable trigger and denominator.

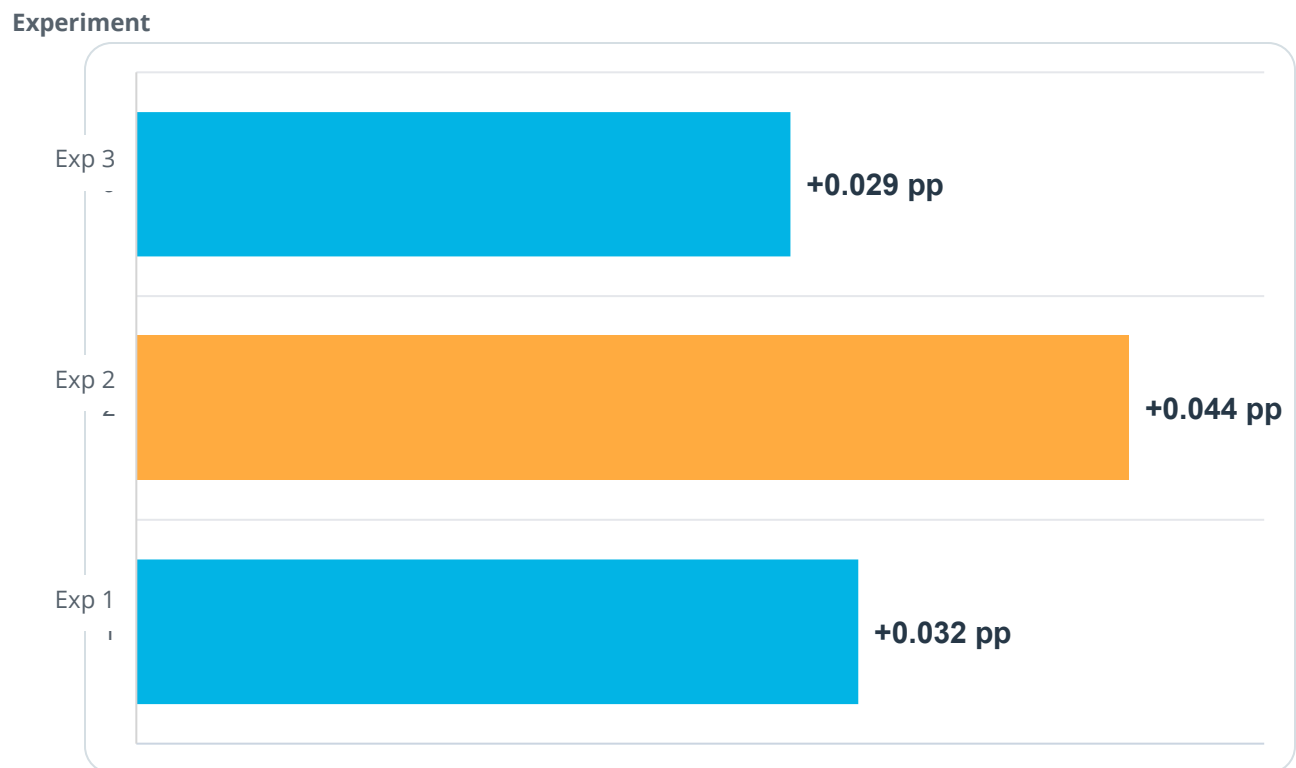
Event	Trigger	Required properties
experiment_exposed	Eligible screen first renders	user_id, session_id, variant, age_band, device
accessibility_used	User invokes feature	variant, feature, input_mode, success
rider_count_selected	Valid rider count confirmed	count, variant, elapsed_ms
search	Valid origin/destination submitted	variant, input_mode, elapsed_ms, error_count
begin_ride	Ride launch confirmed	variant, fare, wait_estimate, payment_status

Tableau Public - 6 views: Experiment Conversion, Experiment Lift, Overall Booking Funnel, Funnel Retention, Age Cohort Conversion and Age Session Mix

<https://public.tableau.com/app/profile/christopher.ohara/viz/FlyberIterativeDesignAnalysis/Sheet3#1>

Observed experiment lift remains inconclusive

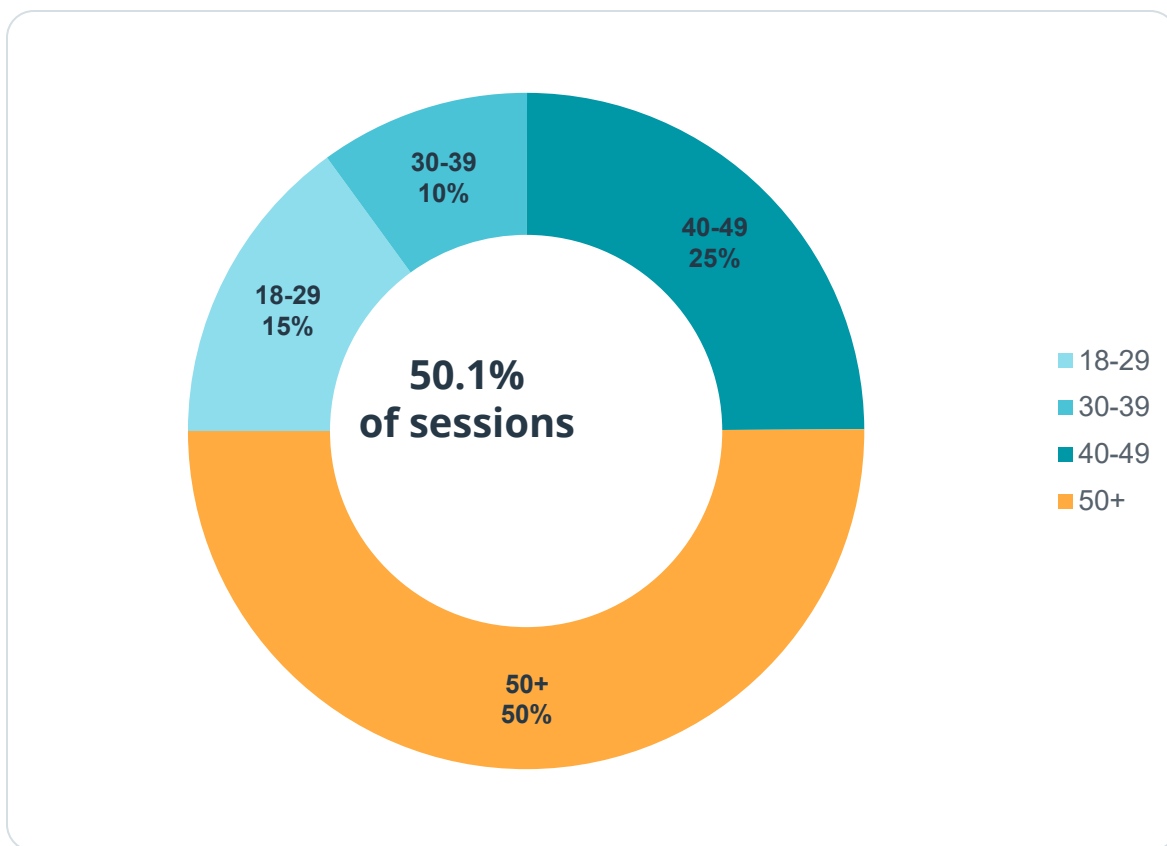
The complementary Tableau lift view shows practical effect size directly. Experiment 2 leads at +0.044 percentage points, but every confidence interval crosses zero and no corrected comparison is significant.



Bar scale: thousandths of a percentage point. Interpret with slide 18: every 95% confidence interval crosses zero; Holm-adjusted p-values range from 0.506 to 0.636.

Age 50+ represents half of observed sessions

The age-session composition view establishes prevalence before interpreting performance. Age 50+ is both the largest mutually exclusive session cohort (50.1%) and the cohort with the largest select-to-search shortfall.

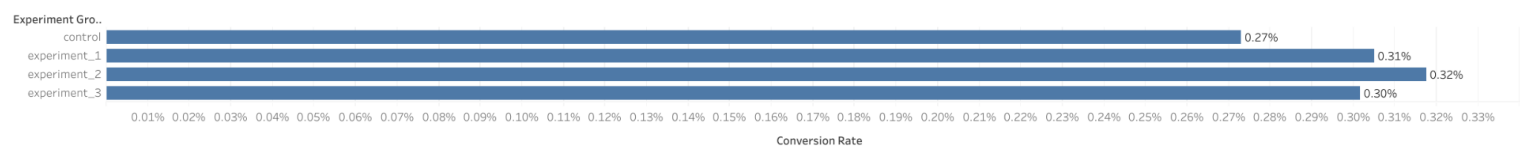


Session counts: 33,878 | 22,760 | 56,256 | 113,261

Experiment Conversion

The public worksheet preserves the original conversion comparison: all three variants are directionally above control, but the observed differences remain small and statistically inconclusive.

Ride Start Rate by Experiment Variant

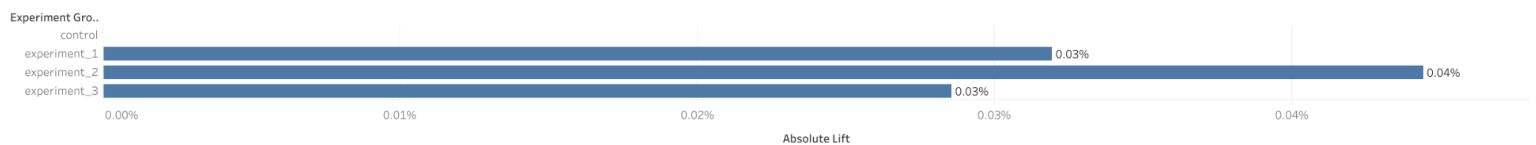


Control 0.27% | Experiment 1 0.31% | Experiment 2 0.32% | Experiment 3 0.30%

Experiment Lift

Absolute lift makes practical effect size explicit. Experiment 2 leads at approximately +0.04 percentage points versus control, but the corrected significance tests still do not support expansion.

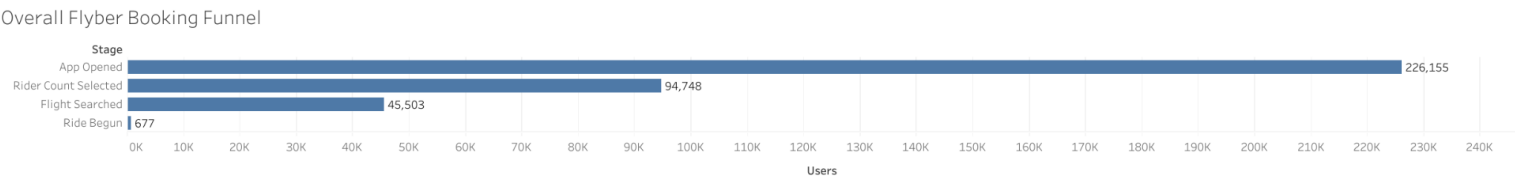
Absolute Ride-Start Lift vs Control



Interpret this effect-size view with slide 18: every 95% confidence interval crosses zero.

Overall Booking Funnel

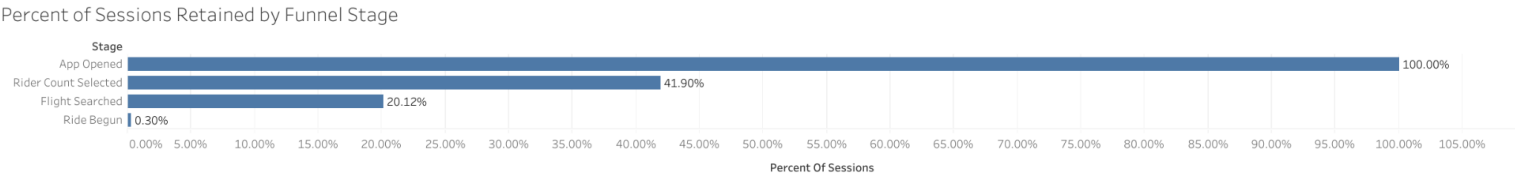
The count view shows where total volume disappears across the ordered journey. Only 677 of 226,155 opened sessions reach ride launch.



226,155 opened -> 94,748 selected riders -> 45,503 searched -> 677 began a ride

Funnel Retention

The proportional view distinguishes retention from raw volume. Rider selection retains 41.9% of opened sessions, search retains 20.1%, and ride launch retains 0.30%.

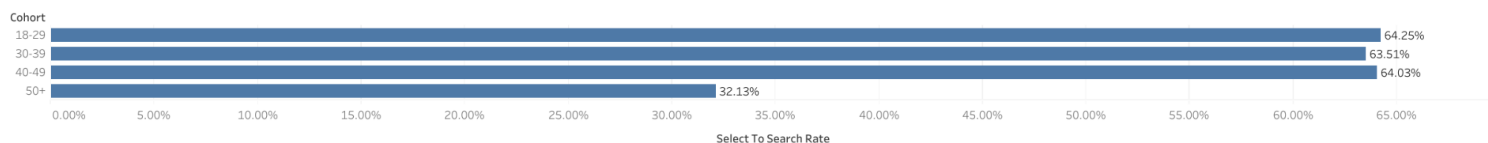


Search to ride is the largest relative loss; open to rider selection is the largest absolute loss.

Age Cohort Conversion

The cohort comparison isolates the primary underperformance: age-50-plus sessions progress from rider selection to search at 32.13%, versus approximately 64% for every younger cohort.

Rider Selection to Search Rate by Age Cohort

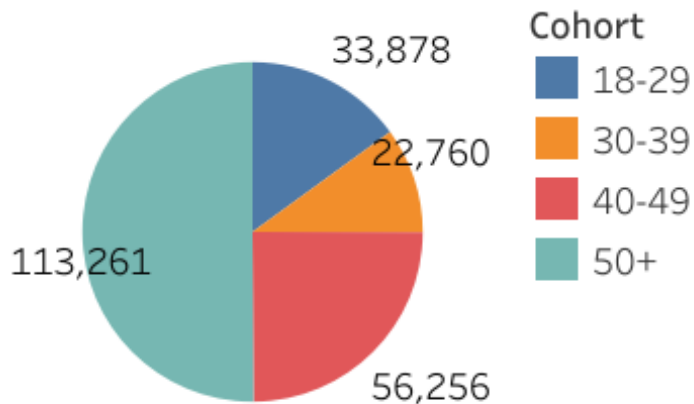


The 50+ select-to-search rate is about half the rate observed among users under age 50.

Age Session Mix

The composition view confirms that the underperforming cohort is not marginal. Age 50+ accounts for 113,261 sessions, or 50.1% of all opened sessions.

Session Mix by
Age Cohort



Because age cohorts are mutually exclusive at the session level, this is a valid part-to-whole view.